

Design of Parallel algorithms

General description

The design of parallel algorithms involves breaking down a problem into smaller, independent sub-problems that can be solved simultaneously on multiple processors. The goal is to improve performance by reducing the total computation time.

Parallel Maximum and Minimum



A parallel algorithm for finding the maximum and the minimal value of numeric vector requires synchronization and communication among threads or process.

Using C++ develop:

1. Design a parallel algorithm
2. An algorithm to find the maximum and Minimum in a vector
3. Write the abstract pseudocode using parallel reduction.
4. How many synchronization steps are required for N elements

Parallel Scans & Prefix Sums

Called as “add-scan” or “prefix sum”. The algorithm adds all the previous values to the current position.

	1	2	3	4	5	6	7	8
A	2	4	1	7	3	0	4	2
P	2	6	7	14	17	17	21	23

1. Design a parallel algorithm to compute the prefix sum

2. Write the abstract pseudocode using a two- phase approach (Upsweep (reduce)) and Downsweep
3. How many synchronization steps are required for N elements

Submit a PDF report detailing the implementations, along with a public repository on GitHub, GitLab, Bitbucket, or a similar platform.